Solar Powered Golf Cart: Testing and Performance Analyses

Jiang Fan
School of Electrical & Electronic Engineering
Singapore Polytechnic
Singapore
e-mail: jiangfan@sp.edu.sg

***Abstract*—An electric golf cart powered by deep cycle, rechargeable lead-acid battery bank is commonly used to carry the golfers and their golf clubs around a golf course at speed not higher than 25km/h. In operation, the cart batteries need to be charged regularly by the mains, which does not only waste energy but also results in short battery lifespan of 1.5 to 3 years. To prolong the lifetime of cart battery, conserve energy and mitigate CO_2 emission, a golf cart can be equipped with solar PV system to harvest solar energy to drive the cart. This paper presents the on-site tests on a 2-seater solar golf cart powered by 420Wp solar PV system and running under tropic weather conditions. Analyses on the testing results show that a solar powered golf cart can not only improve the lifespan of cart battery but also conserve energy and reduce CO_2 emission.**

Keywords-solar photovoltaic; battery storage; electrical cart and energy conservation formatting

I. INTRODUCTION

A golf cart (also named as golf car or buggy) is a small vehicle designed originally to carry the golfers and their golf clubs around a golf course with less effort than walking. Now golf carts come in a wide range of formats and are more generally used to convey small numbers of passengers in short distances at speeds not higher than 25 km/h. Most of golf carts are electrically powered by rechargeable batteries due to their lack of pollutants, lack of noise, and safety for pedestrians and other vehicles although gasoline-powered carts are found in some applications.

Normally an electric golf cart is equipped with 38V or 48V, rechargeable, deep cycle, Lead-acid battery bank. The choice of Lead-acid battery can be flooded or AGM or Gel type. When battery drives a cart and its power fades, the cart's speed becomes slower. Hence the batteries must be charged regularly from the main supply to ensure that the cart can continue to operate. The common life expectancy of a golf cart battery varies with many factors such as storage, temperatures, discharging and charging practices. Generally speaking, the deeper the battery discharges and is charged, the shorter its lifetime will be[1-3]. To prolong the lifetime and reduce energy consumption of the cart's battery, a cart can be equipped with solar PV system to harvest solar energy for driving power. This paper presents the on-site tests of a 2-seater golf cart equipped with $420W_p$ solar PV system. Two days tests conducted for the solar cart were scheduled together with the other golfers' carts. Analyses on the testing results reveal that a solar golf cart can extend the lifespan of cart batteries and also conserve energy and reduce CO_2 emission in a tropic region.

II. THE 2-SEATER SOLAR POWERED GOLF CART AND THE MEASUREMENT SYSTEM FOR THE SITE TESTS

The 2-seater marshal golf cart under testing as shown in Fig. 1 is provided by Singapore Marina Bay Golf club and the solar PV system is installed by Sanyo Asia Pte Ltd. The existing driving system of the cart consists of batteries, micro-computer based energy management unit, DC motor and other devices as detailed in figure 2. The cart motor is a 48V, 3kW DC motor that is driven by 4 U.S. Battery 155AH/C20 12V XC deep cycle AGM rechargeable batteries[4]. To harness the solar energy to operate the cart, two $210W_p$ Sanyo HIT solar PV modules were installed and connected in parallel to charge the batteries through a charge controller and drive the cart motor. The technical specifications of solar PV module and charge controller are listed in tables 1 and 2 respectively.

To investigate the operational performance of solar powered cart accurately during the tests, the 7 sensors mounted on the cart(Fig.2) to acquire the following parameters,

a). Solar irradiance measured by pyranometer;
b). 'I_1', current output of solar PV modules measured by Kyoritsu DC clamp meter;
c). 'V_1', voltage output of solar PV modules measured by resistive voltage divider;
d). 'I_2', current output of charge controller measured by Kyoritsu DC clamp meter;
e). 'I_3', current flowing to the Motor measured by Fluke DC clamp meter;
f). 'V_2', voltage of battery bank measured by resistive voltage divider.

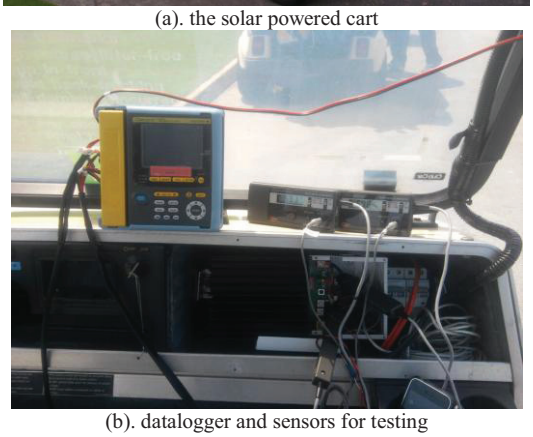
To maintain measurement accuracy in the tests, the 16 bits YOKOGAWA XL-122 data station and the high accuracy DC current clamp meters were selected to sample and record the experimental data in the tests. As depicted in Figs. 1 and 2, the six input channels of the DATA logger are connected to three current clamp meters, two voltage dividers and a pyranometer respectively. The pyranometer was installed on top of the solar buggy to monitor the solar irradiance. Two voltage dividers having ratio of 1 to 3 were used to step down the voltage inputs from solar PV modules and battery bank to comply with the voltage input range of data logger from 100mV to 50V.

978-1-5386-5236-7/18/$31.00 ©2018 IEEE

The basic electrical specifications of the current transducers used for the testing include

- Two AC/DC Digital Clamp Meter with output cord (Kyoritsu Electrical: Model 2010):
  DC Current: Input(0 ~ 20.00A), Accuracy: +/- (2% + 0.5mV)
- One AC/DC digital clamp meter with output cord(Fluke: Model i310s):
  DC Current: input(0 ~ 450A),
  Accuracy: +/- 1% of reading +/-300mA.

(a). the solar powered cart



(b). datalogger and sensors for testing

Figure 1. 2-seater solar powered golf cart equipped with data acquisition system.

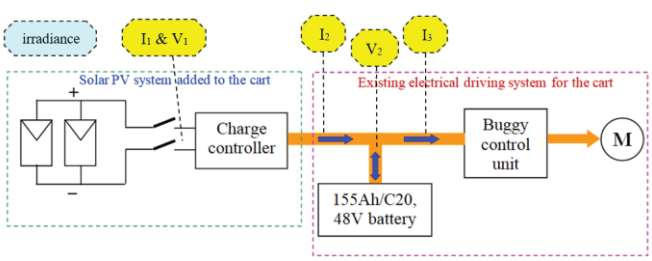


Figure 2. Layout of cart's electrical system and signal sensors for the tests.

TABLE I. DATA OF PV MODULE UNDER STC*

Characteristics	*Normal values*	*Tolerances*
Maximum Power (P_{max})	210W	+10% to -5%
Open circuit voltage (V_{oc})	50.9V	± 10%
Short circuit current (I_{sc})	5.57A	90% or more
Maximum power voltage (V_{pmax})	41.3V	
Maximum power current (I_{pmax})	5.09A	
Max. system open circuit voltage	600 V	

'*': 1000W/m^2 irradiance, AM 1.5, 25 °C cell temperature

TABLE II. SPECIFICATIONS OF CHARGE CONTROLLER: TRI-STAR TS-MPPT-45

Parameters	*Values*
Nominal System Voltage	48 V_{dc}
Maximum Battery Current	45 A_{dc}
Maximum Solar Input Voltage	150 V_{dc}
Battery Operating Voltage Range	8 - 72 V_{dc}
Nominal Maximum Input Power	48V, 2400W
Battery Charging algorithm	4-stage

III. EXPERIMENTS ON SOLAR GOLF CART AT THE GOLF COURSE

According to the arrangement of the golf course, there are normally two half-a-day 18 holes golf games in a day, each game taking average 3 to 4 hours. So a golf cart runs in 3 sessions a day, i.e. (1). the morning session(3~4 hours), (2). the parking session for battery charging(1~2 hours) and (3). the afternoon session(3~4 hours). Two full days on-site tests were scheduled in two separate days to follow the golfers' cart running in the golf course and collect the operating data of the solar cart. During the two days' tests, the main load of solar cart included two persons, one fully loaded golf bag and one water tank filled with water to simulate the weight of another golf bag as shown in Fig. 3.

Figure 3. The solar cart following other golfers' carts to conduct the site test.

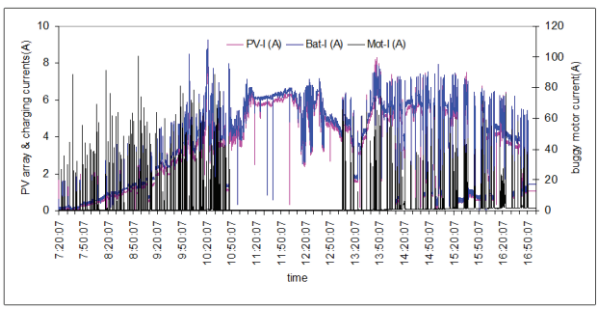
When the morning session was finished, the solar cart was parked under the sunlight to charge the battery for the

afternoon session, which replaced the mains supply charging. The time for waiting period between morning game and afternoon game varies from 1 to 2 hours.

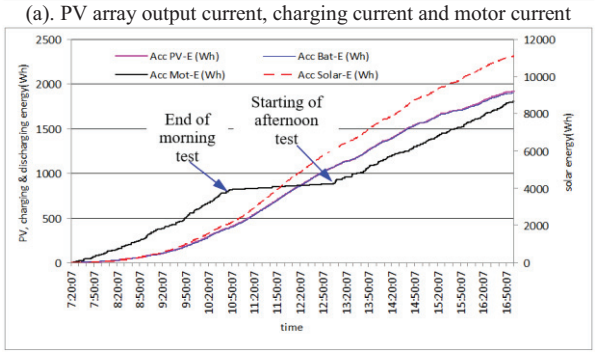
To monitor the dynamic performance of solar cart, the sampling rate of datalogger was set at 1 sample/second and the measurement was conducted continuously from 7.20am to 5pm.

IV. Analyese on The Data And Performance of Solar Cart

The measurement results from two days site-tests are presented in Figs. 4 and 5. The irradiation on day 1 in Fig.4 was 4.32kh/m^2/day, while the irradiation on day 2 in Fig. 5 was 2.81kWh/m^2/day. The tests under different weather conditions in two days help to investigate the performance of the solar cart practically.



(a). PV array output current, charging current and motor current



(b). Solar energy input, PV array output energy, battery charging energy and motor current

Figure 4. On-site experimental results in day 1.

The currents shown in Figs. 4a and 5a show that the maximum output current of PV array is 8.6A, the charging current of charge controller 9.25A and the highest current of motor 91.1A. Although the load current of cart motor was much higher than the charging current of the solar PV system, the energy consumption of cart motor was not very high due to the short intervals of motor operation depending on the golfers' use, i.e. different handicaps. The testing results reveal the daily energy consumed by the cart motor range from 1.8kWh/day to 2.0kWh/day. In comparison, the electrical energy converted from the sunlight was quite consistent when the cart was running under the sunlight.

Figure 4 presents the testing results in day 1. It can be seen that

1). in the morning session from 7:20am to 10:53am, the cart motor consumed more energy than the energy produced by the PV system,

2). in the parking period 10:53am to 1:03pm, the cart batteries gained and stored energy generated by the PV system

3). in afternoon period from 1:03pm to 5:00pm, the energy from solar PV system and battery bank surpassed the energy demand of cart.

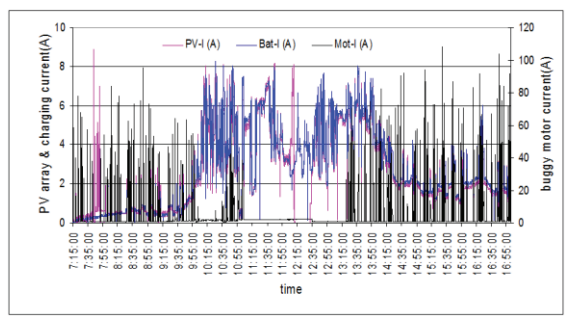
Thanks to the strong irradiation on that day, the total energy produced by solar PV system reached 1.9kWh which was higher than energy demand of the cart, 1.8kWh, in that day.

Similarly, figure 5 shows the experimental results in day 2 that was a cloudy day. It can be seen that

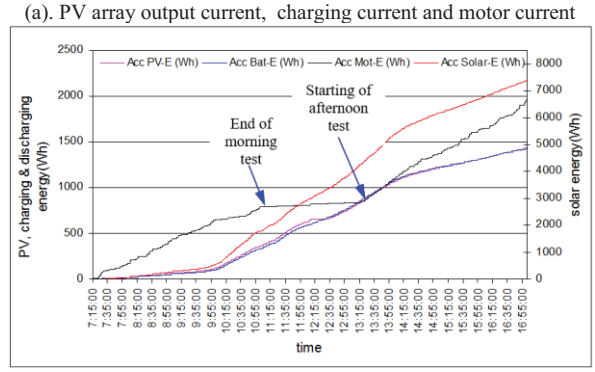
1). in the morning test from 7:15am to 11:04am, the cart was mainly powered by the battery and partly supplied by PV system after 10am;

2). in the parking period from 11:04am to 1:20pm, the PV system charged the battery bank and

3). in the afternoon session from 1:03pm to 5:00pm, the cart motor was driven by PV system and battery bank.



(a). PV array output current, charging current and motor current



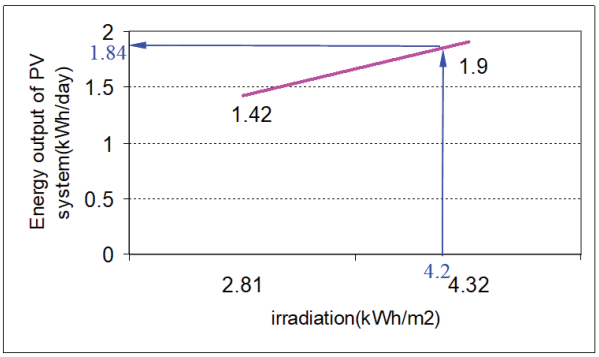
(b). Solar energy input, PV array output energy, battery charging energy and motor current

Figure 5. On-site experimental results in day 2.

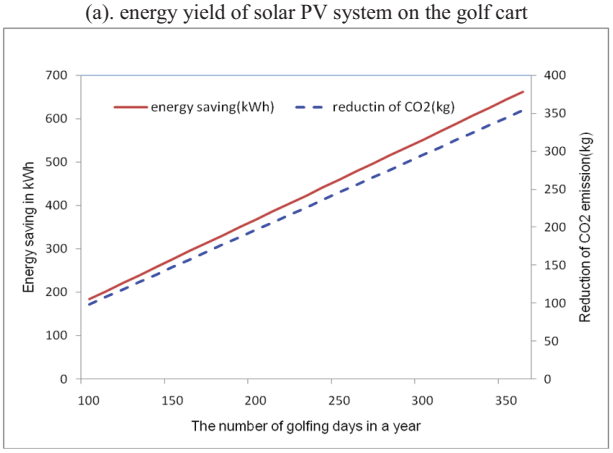
It is apparent to find that the energy used by the cart exceeded the energy produced by solar PV system in whole day due to the cloudy and low irradiance weather. The total energy produced by solar PV system was 1.42kWh which was lower than the total consumption of the cart, 1.96kWh, in that day.

Since the energy output of PV system is proportional to the incoming solar energy as shown in Fig. 6a, the PV

system mounted on the cart can produce about 1.84kWh/day under the 4.2 peak sun hour which is accepted in Singapore[5]. It can conclude that the electrical energy produced by the PV system is close to the daily load demand for the golfers to complete their games. Based on the experimental results and average irradiation of 4.2kWh/m^2/day, we are able to work out energy saving and the reduction of CO_2 for the golf cart in a year as given in Fig. 6b, which will be dependent upon how many days the cart runs in a year.



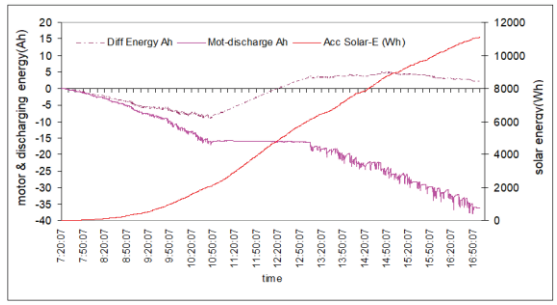
(a). energy yield of solar PV system on the golf cart



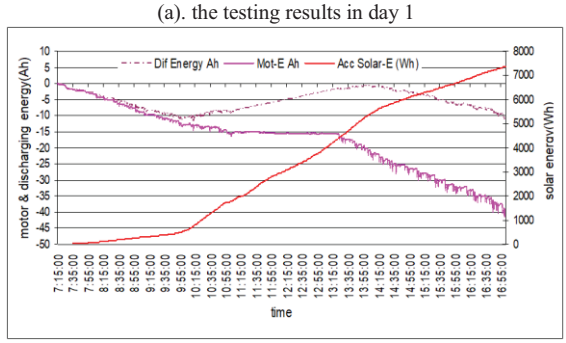
(b). energy saving and reduction of CO_2 for the solar cart

Figure 6. Analyses of PV system conversion efficiency, energy saving & CO_2 reduction.

The life-span of cart batteries is normally quite short from 1.5 to 3 years due to the heavy duty and frequent high current charging operation. The installation of solar PV system on a golf cart helps to improve the operation and lifespan of cart batteries by reducing the DOD of batteries. Fig. 7 presents the energy used by the cart motor and the energy flow of battery bank. The directions of the energy flow of battery bank are defined in such a way that the discharging current is negative and the charging current positive. It is noticeable that the total Ahs consumed by the cart under testing in two different days are very close to the same level, i.e. 40Ah. So for a normal cart without solar power, its battery bank needs to supply 40Ah energy to run the cart. Referring to the parameters provided by battery manufacturer, the depth of discharge(DOD) for a normal cart is about 25.8%. Observing the battery discharging Ah of two experiments on site in Fig. 7, it can be found that the deepest discharges of battery bank in day 1 and day 2 are 8.2Ah and 11.3Ah respectively. So the DOD of battery bank for the solar powered cart dropped to 5.2% and 7.3% of full battery rating.



(a). the testing results in day 1



(b). the testing results in day 2

Figure 7. Energy flows for motor driving and battery discharge.

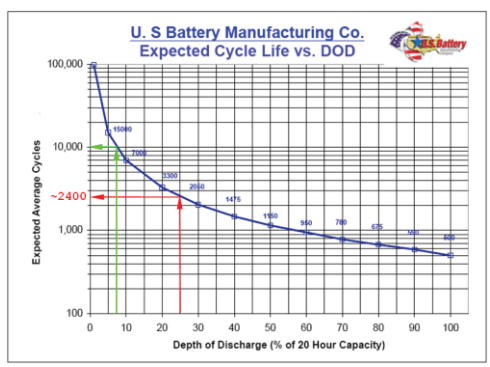


Figure 8. Relationship of battery operational cycle vs. DOD given by U.S. battery.

Comparing life-cycle of battery with its DOD between solar cart and normal cart in Fig. 8, the battery life cycle for a normal cart is about 2400 cycles, while the battery life cycle for the solar powered cart reaches 10000 cycles, about 4 times lifespan of the batteries of a normal cart. It should be noted that the actual lifetime of battery is shorter than the data in figure 8 due to the high discharging current to run the cart and operation at high temperature [6-7].

V. CONCLUSIONS

A solar powered 2-seater golf cart has been tested in 2 different days at the golf course. The testing conditions are

very close to the practical operation of normal golf cart run by the golfers. Analyses on the data presented in this paper reveal that the $420W_p$ solar PV system installed on a golf cart can produce enough electrical energy to power the buggy in a day under the tropical weather conditions. The life-time of batteries for a solar cart can be more than 5 times longer than that of the same batteries used in a normal cart. A solar cart does not only help to save energy and prolong the life time of batteries but also cut down the CO_2 emission.

It can be concluded that a solar powered cart is technically viable, economically feasible and environmental friendly. As the price of solar PV module drops and the electrical tariff increases, the payback for installation of PV system on a cart will be undoubtedly shorter.

ACKNOWLEDGMENT

Author likes to express his thanks to Singapore Polytechnic, Sanyo Asia Pte Ltd and Singapore Marina Bay Golf club for their support to the project.

REFERENCES

[1] Giglioli R, Pelacchi P, Raugi M, Zini G 1988 *A state of charge observer for lead-acid batteries*. Energia Elettrica 65(1):27–33

[2] D.Jaya Deepti, V. Ramanarayanan 2006 *State of Charge of Lead Acid Batter*y, Proceedings of India International Conference on Power Electronics

[3] Corrigan D, Masias A 2011 *Batteries for electric and hybrid vehicles*. Reddy TB (ed) Linden's handbook of batteries, 4th edition. McGraw Hill, New York p15

[4] Care and Maintenance: US battery: http://usbattery.com/info-center/care-and-maintenance/

[5] Fan Jiang 2007 *Investigation of Solar Energy For Photovoltaic Application In Singapore*, proceedings of International Power Engineering Conference (IPEC'2007), Singapore.

[6] Toronto and Region Conservation Authority, 2010 *Evaluation of Solar Assisted, Electric and Gas Golf Carts*.

[7] Dirk Uwe Sauer, Heinz Wenzl 2008 *Comparison of different approaches for lifetime prediction of electrochemical systems-Using lead-acid batteries as example,* Journal of Power Sources, Vol. 176, NO. 2.